

20261014 Quiz

for (MATH200) Differential Equations

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Problem 7.5.5. Find the general solution of the system

$$\mathbf{x}' = \begin{pmatrix} 3 & -1 \\ 6 & -2 \end{pmatrix} \mathbf{x}.$$

Solution. Let

$$\mathbf{A} = \begin{pmatrix} 3 & -1 \\ 6 & -2 \end{pmatrix}.$$

The characteristic equation is

$$\det(\lambda \mathbf{I}_2 - \mathbf{A}) = \lambda^2 - \lambda = \lambda(\lambda - 1) = 0.$$

For the eigenvalue $\lambda_1 = 0$, solving $\mathbf{A}\mathbf{v}_1 = \mathbf{0}$ gives

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

For $\lambda_2 = 1$, solving $(\mathbf{A} - \mathbf{I}_2)\mathbf{v}_2 = \mathbf{0}$ gives

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Since the eigenvalues are distinct, the corresponding eigenvectors are linearly independent. Therefore the general solution is

$$\boxed{\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix}}.$$

The first term is constant because $e^{0t} = 1$.

□

Problem 7.5.11. Solve the initial value problem

$$\mathbf{x}' = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Describe the behavior of the solution as $t \rightarrow \infty$.

Solution. The characteristic equation of the coefficient matrix is

$$\det(\lambda \mathbf{I}_2 - \mathbf{A}) = \lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4) = 0.$$

The eigenvalues $\lambda_1 = 2$ and $\lambda_2 = 4$ have corresponding eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Thus

$$\mathbf{x}(t) = c_1 e^{2t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 e^{4t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The initial condition gives

$$c_1 + c_2 = 3, \quad 3c_1 + c_2 = -1,$$

so $c_1 = -2$ and $c_2 = 5$. Hence

$$\boxed{\mathbf{x}(t) = -2e^{2t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + 5e^{4t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}}.$$

As $t \rightarrow \infty$, the e^{4t} term dominates:

$$e^{-4t} \mathbf{x}(t) \rightarrow 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Thus both components tend to $+\infty$, and the trajectory moves away from the origin, asymptotically in the direction $(1, 1)^T$.

□

Problem 7.5.12. Solve the initial value problem

$$\mathbf{x}' = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}.$$

Describe the behavior of the solution as $t \rightarrow \infty$.

Solution. For the coefficient matrix $\mathbf{A}$, the characteristic equation is

$$\det(\lambda \mathbf{I}_3 - \mathbf{A}) = (\lambda - 1)(\lambda - 2)(\lambda - 3) = 0.$$

Solving $(\mathbf{A} - \lambda_i \mathbf{I}_3) \mathbf{v}_i = \mathbf{0}$ for $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 3$ gives

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

Since the three eigenvalues are distinct, these eigenvectors form a basis of $\mathbb{R}^3$. The general solution is

$$\mathbf{x}(t) = c_1 e^t \mathbf{v}_1 + c_2 e^{2t} \mathbf{v}_2 + c_3 e^{3t} \mathbf{v}_3.$$

Applying the initial condition yields

$$c_2 + 2c_3 = 2, \quad -2c_1 + c_2 + 2c_3 = 0, \quad c_1 + c_3 = 3.$$

Thus $c_1 = 1$, $c_2 = -2$, and $c_3 = 2$, giving

$$\mathbf{x}(t) = e^t \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} - 2e^{2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + 2e^{3t} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

The term with the largest eigenvalue dominates as $t \rightarrow \infty$:

$$e^{-3t}\mathbf{x}(t) \longrightarrow 2 \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

Therefore all three components tend to $+\infty$, and the trajectory moves away from the origin, asymptotically in the direction $(2, 2, 1)^\top$.

□

Problem 7.6.4. Consider the system

$$\mathbf{x}' = \begin{pmatrix} 3 & 4 \\ -2 & -1 \end{pmatrix} \mathbf{x}.$$

- (a) Express the general solution of the given system in terms of real-valued functions.
- (b) Describe the behavior of the solutions as $t \rightarrow \infty$.

Solution. (a) The characteristic equation is

$$\det(\lambda \mathbf{I}_2 - \mathbf{A}) = \lambda^2 - 2\lambda + 5 = 0,$$

with roots $\lambda = 1 \pm 2i$. For $\lambda = 1 + 2i$, an eigenvector is

$$\mathbf{v} = \begin{pmatrix} -1 - i \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} + i \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

By Euler's formula, the real and imaginary parts of the complex solution $e^{(1+2i)t}\mathbf{v}$ are

$$\begin{aligned} \mathbf{x}^{(1)}(t) &= e^t \begin{pmatrix} \sin(2t) - \cos(2t) \\ \cos(2t) \end{pmatrix}, \\ \mathbf{x}^{(2)}(t) &= e^t \begin{pmatrix} -\sin(2t) - \cos(2t) \\ \sin(2t) \end{pmatrix}. \end{aligned}$$

Both are real-valued solutions, and

$$\det(\mathbf{x}^{(1)}(t) \ \mathbf{x}^{(2)}(t)) = e^{2t} \neq 0.$$

Hence they form a fundamental set and the real-valued general solution is

$$\mathbf{x}(t) = c_1 e^t \begin{pmatrix} \sin(2t) - \cos(2t) \\ \cos(2t) \end{pmatrix} + c_2 e^t \begin{pmatrix} -\sin(2t) - \cos(2t) \\ \sin(2t) \end{pmatrix}.$$

(b) The eigenvalues have positive real part, so every nonzero solution spirals away from the origin as $t \rightarrow \infty$. The rotation is clockwise because

$$x_1 x_2' - x_2 x_1' = -2((x_1 + x_2)^2 + x_2^2) < 0 \quad (\mathbf{x} \neq \mathbf{0}).$$

Hence the origin is an unstable clockwise spiral point; the zero solution remains at the origin. $\square$

Problem 7.7.2. For the system

$$\mathbf{x}' = \begin{pmatrix} 2 & 1 \\ -5 & -2 \end{pmatrix} \mathbf{x},$$

- (a) Find a fundamental matrix.
- (b) Find the fundamental matrix $\Phi(t)$ satisfying $\Phi(0) = \mathbf{I}_2$.

Solution. (a) The characteristic equation is

$$\det(\lambda \mathbf{I}_2 - \mathbf{A}) = \lambda^2 + 1 = 0,$$

so $\lambda = \pm i$. An eigenvector for $\lambda = i$ is

$$\mathbf{v} = \begin{pmatrix} 1 \\ -2 + i \end{pmatrix}.$$

Taking the real and imaginary parts of $e^{it}\mathbf{v}$ gives

$$\mathbf{x}^{(1)}(t) = \begin{pmatrix} \cos t \\ -2 \cos t - \sin t \end{pmatrix}, \quad \mathbf{x}^{(2)}(t) = \begin{pmatrix} \sin t \\ \cos t - 2 \sin t \end{pmatrix}.$$

Placing these independent solutions in the columns gives

$$\Psi(t) = \begin{pmatrix} \cos t & \sin t \\ -2 \cos t - \sin t & \cos t - 2 \sin t \end{pmatrix}.$$

Indeed, $\det \Psi(t) = 1 \neq 0$, so Ψ is a fundamental matrix.

(b) To impose the normalization at $t = 0$, use

$$\Phi(t) = \Psi(t)\Psi(0)^{-1}.$$

Since

$$\Psi(0) = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}, \quad \Psi(0)^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix},$$

we obtain

$$\Phi(t) = \begin{pmatrix} \cos t + 2 \sin t & \sin t \\ -5 \sin t & \cos t - 2 \sin t \end{pmatrix}.$$

Direct substitution confirms

$$\Phi'(t) = \mathbf{A}\Phi(t), \quad \Phi(0) = \mathbf{I}_2,$$

as required. $\square$